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publications & preprints

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Manuscripts Under Review

7. Tongtong Liang, Siqi Kou, Ziqiao Xi, Esha Singh, Kun Zhou, Zhijie Deng, Alexander Cloninger, Yu-Xiang Wang, and **Rahul Parhi**. "Residual-Stream Burden Shapes Representation Learning in Diffusion Transformers". Sept. 2026. arXiv: 2609.33895 [cs.CV].

6. Henry Pritchard and **Rahul Parhi**. "Plug-and-Play Methods Provably Converge Even with Improperly Trained Denoisers: Convergence by Architectural Design". Sept. 2026. arXiv: 2609.33087 [math.OC].

5. Soroush Mesforush and **Rahul Parhi**. "Lasso Universality Under Linearly Dependent Covariates in the Sparse Regime". Aug. 2026. arXiv: 2608.08390 [math.ST].

4. Greg Ongie and **Rahul Parhi**. "Representation Costs in Data Science: Foundations and the Quasi-Banach Spaces of Deep Neural Networks". June 2026. arXiv: 2606.14954 [math.FA].

3. Anıl Kamber and **Rahul Parhi**. "On the Loss Landscape Geometry of Regularized Deep Matrix Factorization: Uniqueness and Sharpness". Mar. 2026. arXiv: 2603.27072 [stat.ML].

2. Ben Adcock, Gregor Maier, and **Rahul Parhi**. "Towards Sharp Minimax Risk Bounds for Operator Learning". Dec. 2025. arXiv: 2512.17805 [math.ST].

1. Ronald DeVore, Robert D. Nowak, **Rahul Parhi**, Guergana Petrova, and Jonathan W. Siegel. "Optimal Recovery Meets Minimax Estimation". Feb. 2025. arXiv: 2502.17671 [math.ST].

Journal Publications

15. Bassam El Rawas, Michael Unser, and **Rahul Parhi**. "Optimal Nonlinear Approximation with Exponential Splines". In: *Journal of Approximation Theory* 321–322.106347 (2027), pp. 1–18. DOI: 10.1016/j.jat.2026.106347.

14. Luke McDermott, Robert W. Heath, and **Rahul Parhi**. "LoLA: Low-Rank Linear Attention with Sparse Caching". In: *Transactions on Machine Learning Research* (2026). ISSN: 2835-8856. URL: https://openreview.net/forum?id=3KhDA252y3.

13. **Rahul Parhi** and Robert D. Nowak. "Compositional Function Spaces for Deep Learning". In: *SIAM Review* 68.1 (2026). **(SIGEST)**, pp. 127–149. DOI: 10.1137/25M1802948.

12. Henry Pritchard and **Rahul Parhi**. "Nonasymptotic Convergence Rates for Plug-and-Play Methods With MMSE Denoisers". In: *IEEE Transactions on Computational Imaging* 12 (2026), pp. 1205–1214. DOI: 10.1109/TCI.2026.3707068.

11. Ronald DeVore, Robert D. Nowak, **Rahul Parhi**, and Jonathan W. Siegel. "Weighted Variation Spaces and Approximation by Shallow ReLU Networks". In: *Applied and Computational Harmonic Analysis* 74.101713 (2025). **(Charles Chui Young Researcher Best Paper Award)**, pp. 1–22. DOI: 10.1016/j.acha.2024.101713.

10. **Rahul Parhi**, Pakshal Bohra, Ayoub El Biari, Mehrsa Pourya, and Michael Unser. "Random ReLU Neural Networks as Non-Gaussian Processes". In: *Journal of Machine Learning Research* 26.19 (2025), pp. 1–31. URL: https://jmlr.org/papers/v26/24-0737.html.

9. **Rahul Parhi** and Michael Unser. "Function-Space Optimality of Neural Architectures with Multivariate Nonlinearities". In: *SIAM Journal on Mathematics of Data Science* 7.1 (2025), pp. 110–135. DOI: 10.1137/23M1620971.

8. **Rahul Parhi** and Michael Unser. "Distributional Extension and Invertibility of the k-Plane Transform and Its Dual". In: *SIAM Journal on Mathematical Analysis* 56.4 (2024), pp. 4662–4686. DOI: 10.1137/23M1556721.

7. Joseph Shenouda, **Rahul Parhi**, Kangwook Lee, and Robert D. Nowak. "Variation Spaces for Multi-Output Neural Networks: Insights on Multi-Task Learning and Network Compression". In: *Journal of Machine Learning Research* 25.231 (2024), pp. 1–40. URL: https://www.jmlr.org/papers/v25/23-0677.html.

6. **Rahul Parhi** and Robert D. Nowak. "Near-Minimax Optimal Estimation With Shallow ReLU Neural Networks". In: *IEEE Transactions on Information Theory* 69.2 (2023), pp. 1125–1140. DOI: 10.1109/TIT.2022.3208653.

5. **Rahul Parhi** and Robert D. Nowak. "Deep Learning Meets Sparse Regularization: A signal processing perspective". In: *IEEE Signal Processing Magazine* 40.6 (2023), pp. 63–74. DOI: 10.1109/MSP.2023.3286988.

4. **Rahul Parhi** and Michael Unser. "The Sparsity of Cycle Spinning for Wavelet-Based Solutions of Linear Inverse Problems". In: *IEEE Signal Processing Letters* 30 (2023), pp. 568–572. DOI: 10.1109/LSP.2023.3275916.

3. **Rahul Parhi** and Robert D. Nowak. "What Kinds of Functions Do Deep Neural Networks Learn? Insights from Variational Spline Theory". In: *SIAM Journal on Mathematics of Data Science* 4.2 (2022). **(SIGEST Award)**, pp. 464–489. DOI: 10.1137/21M1418642.

2. **Rahul Parhi** and Robert D. Nowak. "Banach Space Representer Theorems for Neural Networks and Ridge Splines". In: *Journal of Machine Learning Research* 22.43 (2021), pp. 1–40. URL: https://jmlr.org/papers/v22/20-583.html.

1. **Rahul Parhi** and Robert D. Nowak. "The Role of Neural Network Activation Functions". In: *IEEE Signal Processing Letters* 27 (2020), pp. 1779–1783. DOI: 10.1109/LSP.2020.3027517.

Journal-Equivalent ML/AI Conference Publications

5. Anıl Kamber and **Rahul Parhi**. "Sharpness of Minima in Deep Matrix Factorization". In: *Advances in Neural Information Processing Systems (NeurIPS)*. **(Spotlight)**. 2026.

4. Tongtong Liang, Alexander Cloninger, **Rahul Parhi**, and Yu-Xiang Wang. "Generalization Below the Edge of Stability: The Role of Data Geometry". In: *International Conference on Learning Representations (ICLR)*. 2026, pp. 72969–73017. URL: https://proceedings.iclr.cc/paper_files/paper/2026/hash/76bfd12a31cff845faec8d082c38fb3a-Abstract-Conference.html.

3. Tongtong Liang, Esha Singh, **Rahul Parhi**, Alexander Cloninger, and Yu-Xiang Wang. "Does Sparse Connectivity Improve Generalization? Convolutional Networks Below the Edge of Stability". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2026.

2. Akash Kumar, **Rahul Parhi**, and Mikhail Belkin. "A Gap Between the Gaussian RKHS and Neural Networks: An Infinite-Center Asymptotic Analysis". In: *Conference on Learning Theory (COLT)*. Vol. 291. 2025, pp. 3463–3485. URL: https://proceedings.mlr.press/v291/kumar25b.html.

1. Tongtong Liang, Dan Qiao, Yu-Xiang Wang, and **Rahul Parhi**. "Stable Minima of ReLU Neural Networks Suffer from the Curse of Dimensionality: The Neural Shattering Phenomenon". In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 38. **(Spotlight)**. 2025, pp. 24686–24750. DOI: 10.52202/085713-0831.

Other Conference Publications

7. **Rahul Parhi** and Ben Adcock. "Upper Bounds on Averaged Sampling Numbers for General Model Classes". In: *International Conference on Sampling Theory and Applications (SampTA)*. 2025, pp. 1–5. DOI: 10.1109/SampTA64769.2025.11133520.

6. Abhinav Uppal, Min Suk Lee, Teng Fei, Adyant Balaji, Zhaoyi Liu, Tzyy-Ping Jung, Lara M. Rangel, Virginia R. De Sa, **Rahul Parhi**, Akshay Paul, Yuchen Xu, and Gert Cauwenberghs. "Brain-Body Coupling in Listening to Metronomic Sounds and Music". In: *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2025, pp. 1–5. DOI: 10.1109/EMBC58623.2025.11251651.

5. **Rahul Parhi** and Michael Unser. "Modulation Spaces and the Curse of Dimensionality". In: *International Conference on Sampling Theory and Applications (SampTA)*. 2023, pp. 1–5. DOI: 10.1109/SampTA59647.2023.10301395.

4. Joseph Shenouda, **Rahul Parhi**, and Robert D. Nowak. "A Continuous Transform for Localized Ridgelets". In: *International Conference on Sampling Theory and Applications (SampTA)*. 2023, pp. 1–5. DOI: 10.1109/SampTA59647.2023.10301398.

3. **Rahul Parhi** and Robert D. Nowak. "On Continuous-Domain Inverse Problems with Sparse Superpositions of Decaying Sinusoids as Solutions". In: *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2022, pp. 5603–5607. DOI: 10.1109/ICASSP43922.2022.9746165.

2. **Rahul Parhi**, Michael Schliep, and Nicholas Hopper. "MP3: A More Efficient Private Presence Protocol". In: *International Conference on Financial Cryptography and Data Security*. 2018, pp. 38–57. DOI: 10.1007/978-3-662-58387-6_3.

1. **Rahul Parhi**, Chris H. Kim, and Keshab K. Parhi. "Fault-Tolerant Ripple-Carry Binary Adder Using Partial Triple Modular Redundancy (PTMR)". In: *IEEE International Symposium on Circuits and Systems (ISCAS)*. 2015, pp. 41–44. DOI: 10.1109/ISCAS.2015.7168565.

Lightly-Refereed Workshop Papers

6. Tongtong Liang, Esha Singh, **Rahul Parhi**, Alex Cloninger, and Yu-Xiang Wang. "What Architectural Inductive Bias Makes Diffusion Models Succeed? A Perspective from the Implicit Regularization of Gradient Descent". In: *ICML Workshop on Foundations of Deep Generative Models: Understanding Memorization, Generalization, and Reasoning*. 2026. URL: https://openreview.net/forum?id=Pl9Qg3s0VQ.

5. Luke McDermott, Robert W. Heath, and **Rahul Parhi**. "Lifelong In-Context Learning with Transformers Requires Parametric Forms of Attention". In: *NeurIPS Workshop on Towards Test-Time Continual Learning Agents (TTCL)*. 2026. URL: https://openreview.net/forum?id=0BL7Z0jrPE.

4. Anıl Kamber and **Rahul Parhi**. "Hessian Spectrum is Constant Across Minimizers in Regularized Deep Scalar Factorization". In: *NeurIPS Workshop on Optimization for Machine Learning (OPT)*. 2025. URL: https://opt-ml.org/papers/2025/paper84.pdf.

3. Tongtong Liang, Alex Cloninger, **Rahul Parhi**, and Yu-Xiang Wang. "Data Geometry Determines Generalization Below the Edge-of-Stability". In: *NeurIPS Workshop on Optimization for Machine Learning (OPT)*. 2025. URL: https://opt-ml.org/papers/2025/paper78.pdf.

2. Luke McDermott and **Rahul Parhi**. "Finding Stable Subnetworks at Initialization with Dataset Distillation". In: *ICLR Workshop on Neural Network Weights as a New Data Modality*. 2025. URL: https://weight-space-learning.github.io/papers.

1. **Rahul Parhi**, Jack Wolf, and Robert D. Nowak. "On the Sparsity of Deep Neural Networks in the Overparameterized Regime: An Empirical Study". In: *ICML Workshop on Overparameterization: Pitfalls & Opportunities (OPPO)*. 2021. URL: https://sites.google.com/view/icml2021oppo/accepted-papers.

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